

最优化理论与方法

第7讲 解析法、梯度下降法、线搜索梯度下降法

詹大为

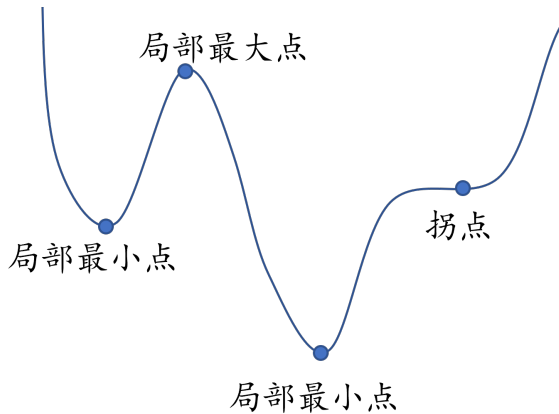
计算机与人工智能学院

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1. 解析法

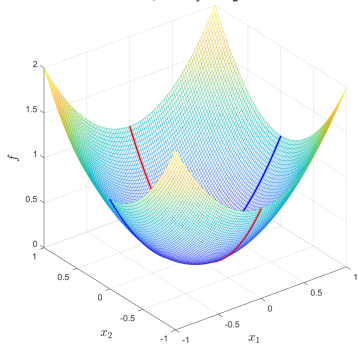
导数为零的点为驻点

$$f'(x) = 0$$

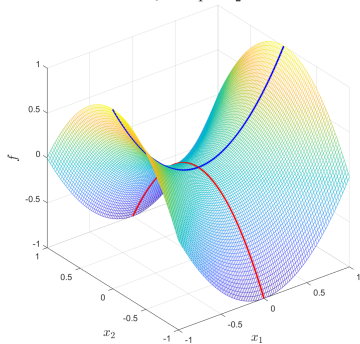


1. 解析法

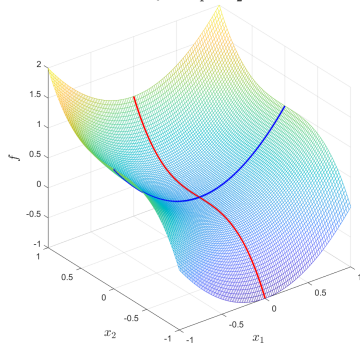
$$f = x_1^2 + x_2^2$$



$$f = x_1^2 - x_2^2$$

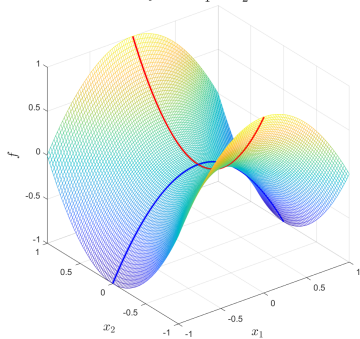


$$f = x_1^2 + x_2^3$$

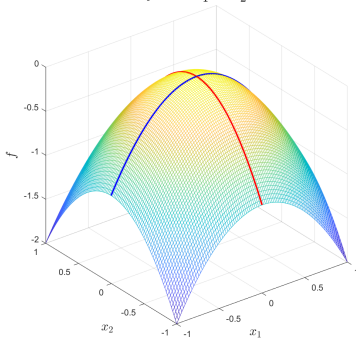


1. 解析法

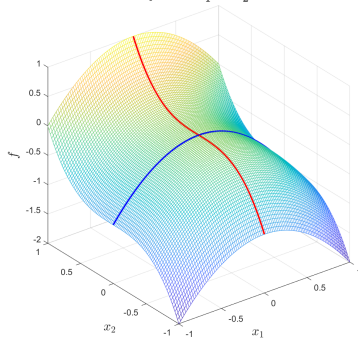
$$f = -x_1^2 + x_2^2$$



$$f = -x_1^2 - x_2^2$$

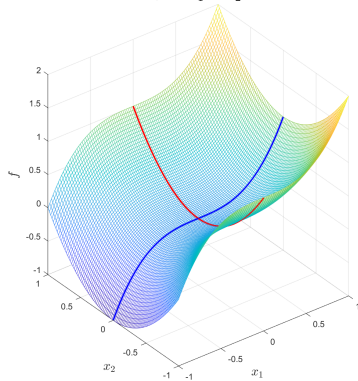


$$f = -x_1^2 + x_2^3$$

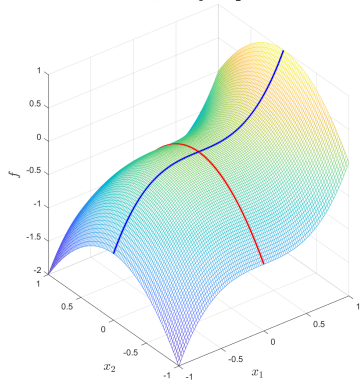


1. 解析法

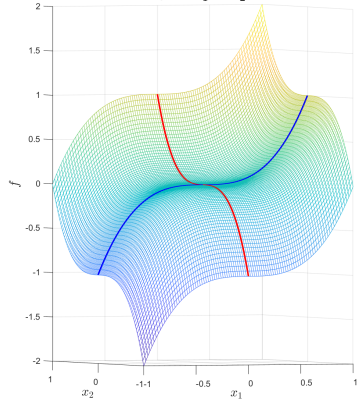
$$f = x_1^3 + x_2^2$$



$$f = x_1^3 - x_2^2$$

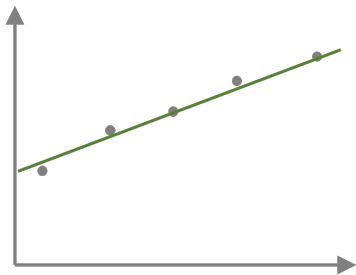


$$f = x_1^3 + x_2^3$$



1. 解析法：示例

x	y	f
1	4.5	$w_1 + w_2$
2	7.1	$2w_1 + w_2$
3	9.9	$3w_1 + w_2$
4	11.9	$4w_1 + w_2$
5	12.3	$5w_1 + w_2$



$$Loss(w_1, w_2) = 55w_1^2 + 5w_2^2 + 30w_1w_2 - 315w_1 - 91.4w_2 + 461.57$$

求损失函数驻点？如何证明这个驻点是全局最小点？

1. 解析法：示例

$$Loss(w_1, w_2) = 55w_1^2 + 5w_2^2 + 30w_1w_2 - 315w_1 - 91.4w_2 + 461.57$$

$$\nabla Loss(w_1, w_2) = \begin{bmatrix} 110w_1 + 30w_2 - 315 \\ 10w_2 + 30w_1 - 91.4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ 解得 } w_1^* = 2.04, w_2^* = 3.02。$$

$$\text{因为 } \nabla^2 Loss(w_1^*, w_2^*) = \begin{bmatrix} 110 & 30 \\ 30 & 10 \end{bmatrix} \text{ 为正定矩阵，驻点为局部最小值点。}$$

$$\text{因为 } \forall \mathbf{w} \in \mathcal{R}^2, \nabla^2 Loss(w_1, w_2) = \begin{bmatrix} 110 & 30 \\ 30 & 10 \end{bmatrix} \text{ 为正定矩阵，所以 } Loss(w_1, w_2) \text{ 为凸函数，局部最小点为全局最小点。}$$

2. 梯度下降法: Gradient Decent

假设 $\boldsymbol{x}^{(1)} \in \{\boldsymbol{x} \mid \|\boldsymbol{x}^{(1)} - \boldsymbol{x}^{(0)}\| \leq \epsilon\}$, 将 $\boldsymbol{x}^{(1)}$ 在 $\boldsymbol{x}^{(0)}$ 进行一阶泰勒展开得:

$$f(\boldsymbol{x}^{(1)}) \approx f(\boldsymbol{x}^{(0)}) + \nabla^T f(\boldsymbol{x}^{(0)}) (\boldsymbol{x}^{(1)} - \boldsymbol{x}^{(0)})$$

希望 $f(\boldsymbol{x}^{(1)}) < f(\boldsymbol{x}^{(0)})$, 则

$$\begin{aligned} f(\boldsymbol{x}^{(0)}) + \nabla^T f(\boldsymbol{x}^{(0)}) (\boldsymbol{x}^{(1)} - \boldsymbol{x}^{(0)}) &< f(\boldsymbol{x}^{(0)}) \\ \nabla^T f(\boldsymbol{x}^{(0)}) (\boldsymbol{x}^{(1)} - \boldsymbol{x}^{(0)}) &< 0 \end{aligned}$$

选取 $\boldsymbol{x}^{(1)} - \boldsymbol{x}^{(0)} = -\eta \nabla f(\boldsymbol{x}^{(0)})$, $\eta > 0$ 满足以上条件, 即

$$\boldsymbol{x}^{(1)} = \boldsymbol{x}^{(0)} - \eta \nabla f(\boldsymbol{x}^{(0)})$$

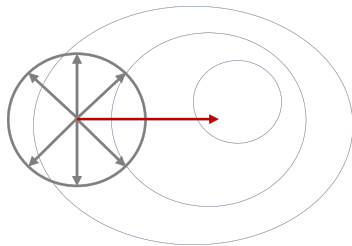
2. 梯度下降法：为什么沿负梯度下降最快？

下降方向 $\Delta \mathbf{x} = \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \end{bmatrix}$ ，负梯度方向 $-\nabla f(\mathbf{x}) = \begin{bmatrix} -\frac{\partial f}{\partial x_1} \\ -\frac{\partial f}{\partial x_2} \end{bmatrix}$

f 下降量

$$\begin{aligned}\Delta f &= -\frac{\partial f}{\partial x_1} \Delta x_1 - \frac{\partial f}{\partial x_2} \Delta x_2 \\ &= -\nabla f(\mathbf{x}) \cdot \Delta \mathbf{x} \\ &= |-\nabla f(\mathbf{x})| \cdot |\Delta \mathbf{x}| \cdot \cos \theta\end{aligned}$$

当 $\theta = 0$ 时， Δf 取最大值。



2. 梯度下降法：初始点与步长的影响

$$\boldsymbol{x}^{(k+1)} = \boldsymbol{x}^{(k)} - \eta \nabla f(\boldsymbol{x}^{(k)})$$

初始点的影响

- (1) 对于凸函数，初始点不影响收敛性，但影响收敛速度。
- (2) 对于非凸函数，不同初始点可能收敛到不同局部最小值点。
- (3) 对于非凸函数，使用多个不同初始点可以增加搜索全局最小值点的概率。

步长的影响

- (1) 步长过小，收敛速度可能比较慢。
- (2) 步长多大，可能发散。
- (3) 使用**线搜索**方法选取步长，或**自适应步长**方法调整步长。

2. 梯度下降法：收敛速度理论

若 $f(\mathbf{x})$ 为连续可导的凸函数，若其梯度函数 Lipschitz 连续， $L > 0$ 为 Lipschitz 常数，当 $\eta \leq \frac{1}{L}$ ，梯度下降可收敛到全局最优解 $\mathbf{x}^*$ ，且收敛速度为线性

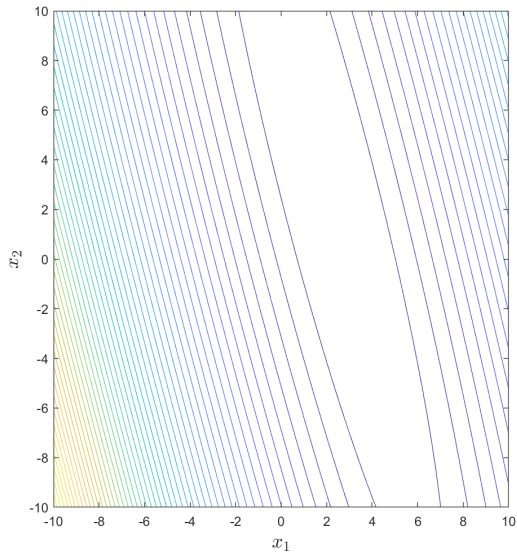
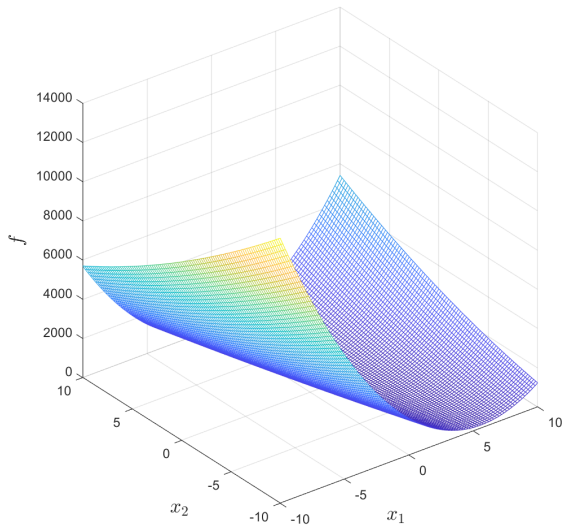
$$f(\mathbf{x}^{(k)}) - f(\mathbf{x}^*) \leq \frac{\|\mathbf{x}^{(0)} - \mathbf{x}^*\|_2^2}{2\eta k}$$

梯度函数 Lipschitz 连续

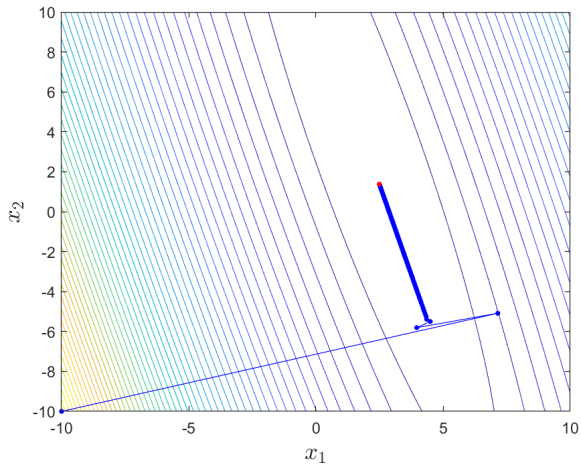
$$\frac{\|\nabla f(\mathbf{x}^{(1)}) - \nabla f(\mathbf{x}^{(2)})\|_2}{\|\mathbf{x}^{(1)} - \mathbf{x}^{(2)}\|_2} \leq L$$

2. 梯度下降法：示例

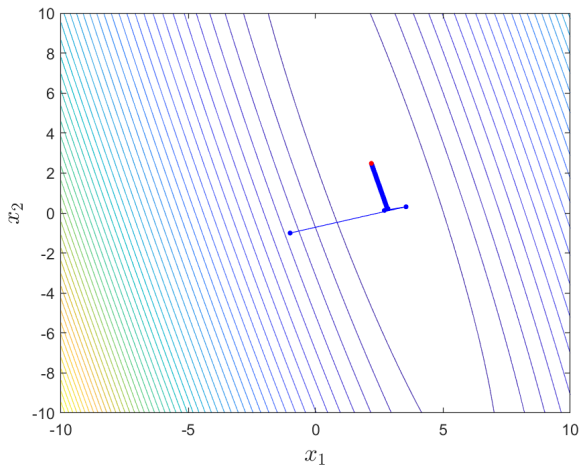
$$f = 55w_1^2 + 5w_2^2 + 30w_1w_2 - 315w_1 - 91.4w_2 + 461.57$$



2. 梯度下降法：示例

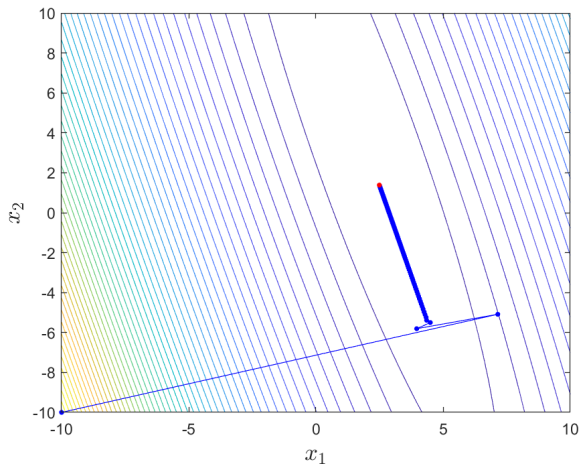


$$\mathbf{x}^{(0)} = [-10, -10]^T, f_{\min} = 4.69$$

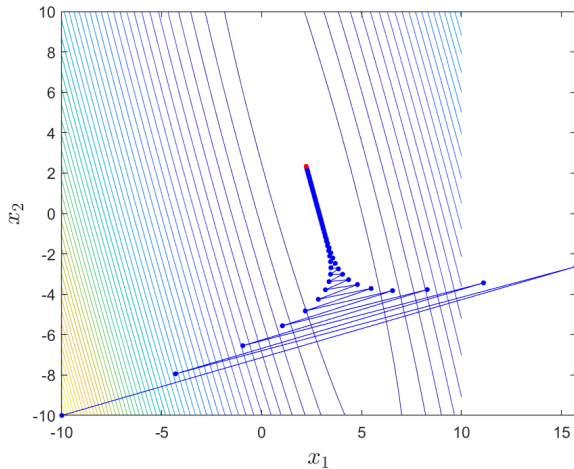


$$\mathbf{x}^{(0)} = [-1, -1]^T, f_{\min} = 2.52$$

2. 梯度下降法：示例



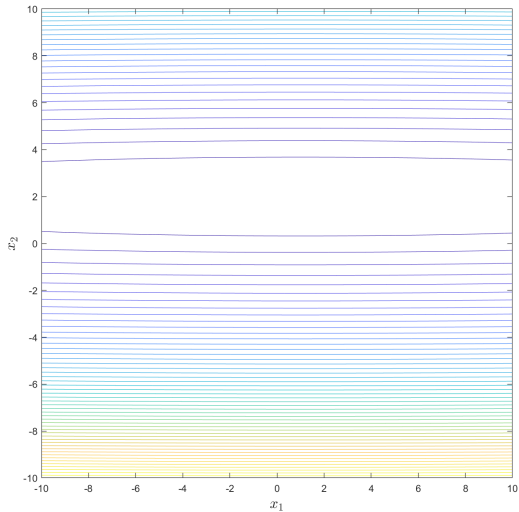
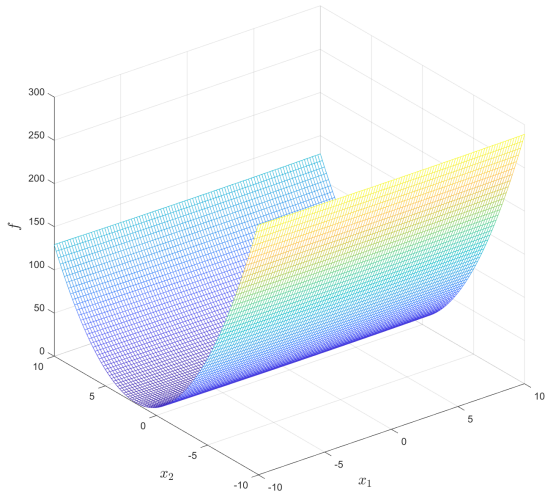
$$\eta = 0.01, f_{\min} = 4.69$$



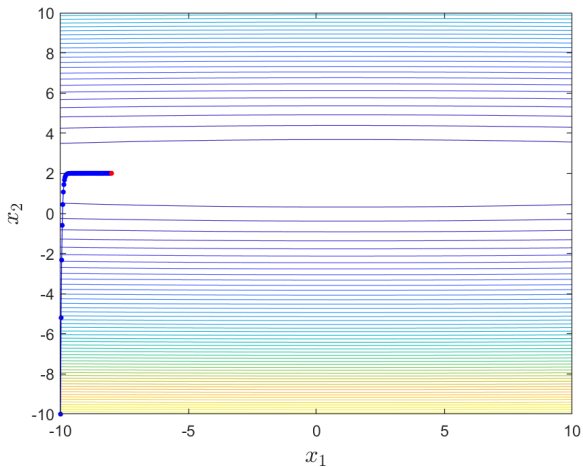
$$\eta = 0.015, f_{\min} = 2.69$$

2. 梯度下降法：示例

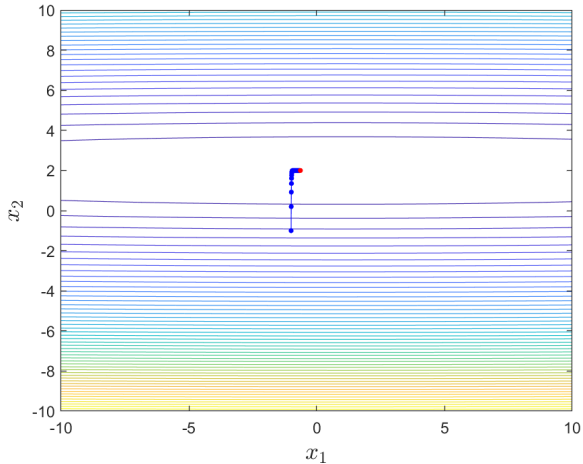
$$f = 0.01(x_1 - 1)^2 + 2(x_2 - 2)^2$$



2. 梯度下降法：示例

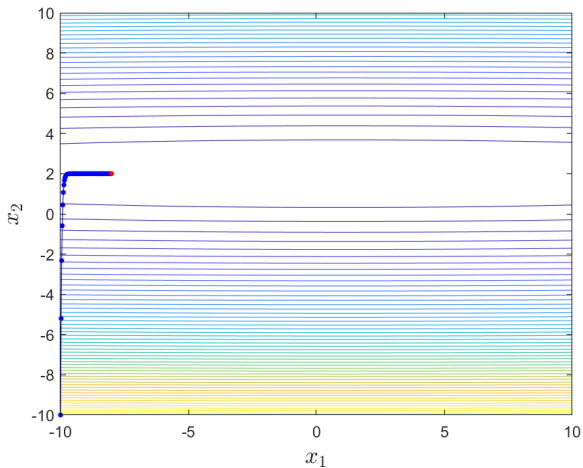


$$\mathbf{x}^{(0)} = [-10, -10]^T, f_{\min} = 0.81$$

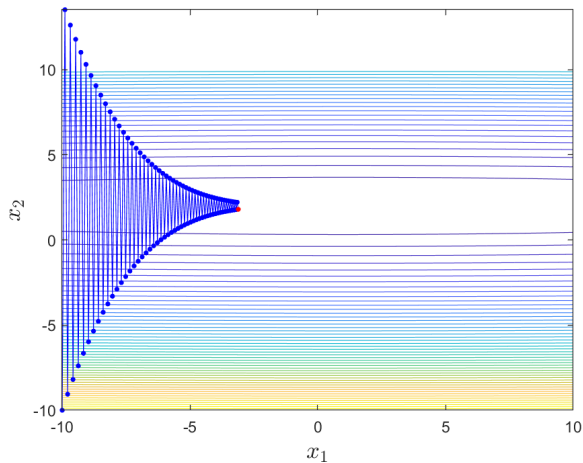


$$\mathbf{x}^{(0)} = [-1, -1]^T, f_{\min} = 0.03$$

2. 梯度下降法：示例

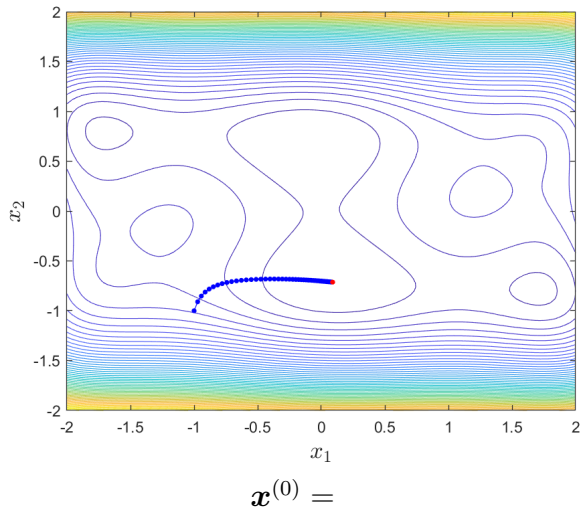
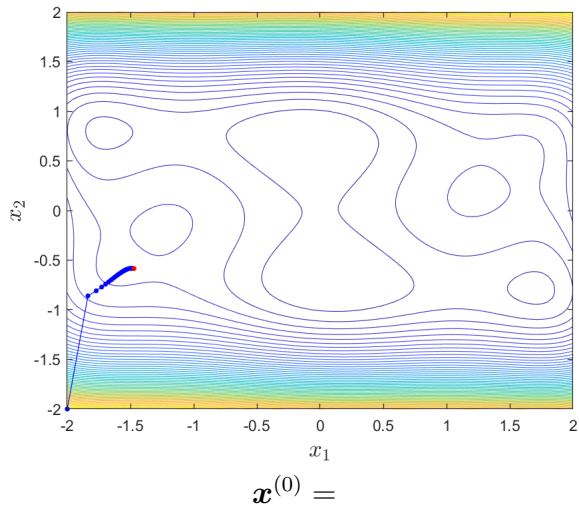


$$\eta = 0.1, f_{\min} = 0.81$$

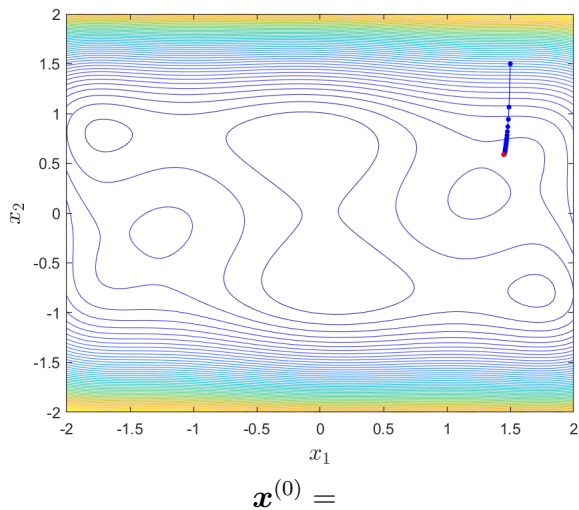
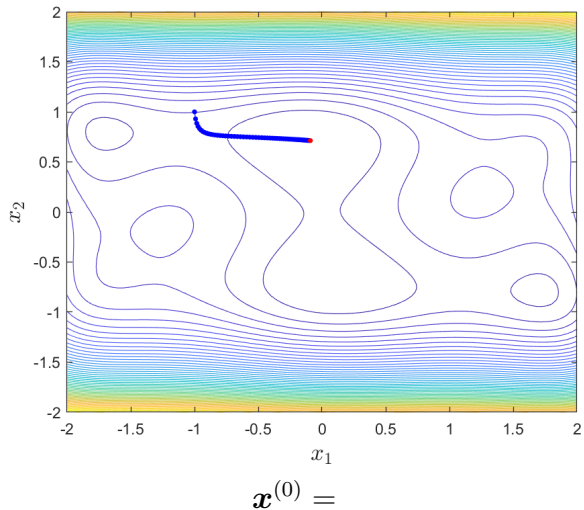


$$\eta = 0.49, f_{\min} = 0.25$$

2. 梯度下降法：非凸函数



2. 梯度下降法：非凸函数



3. 线搜索步长梯度下降: Line Search

$$\boldsymbol{x}^{(k+1)} = \boldsymbol{x}^{(k)} - \eta \nabla f(\boldsymbol{x}^{(k)})$$

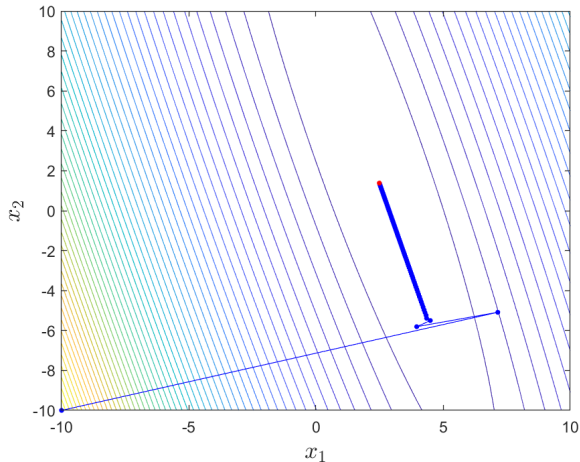
研究 $\boldsymbol{x}^{(k+1)}$ 的目标值

$$f(\boldsymbol{x}^{(k+1)}) = f(\boldsymbol{x}^{(k)} - \eta \nabla f(\boldsymbol{x}^{(k)})) = f(\eta)$$

此时最优步长为

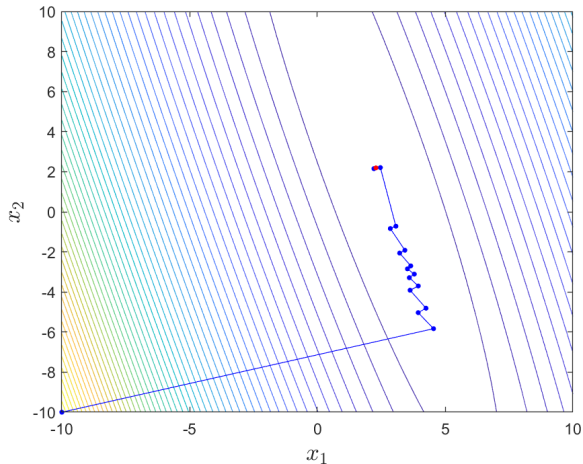
$$\eta^* = \arg \min f(\eta) = \arg \min f(\boldsymbol{x}^{(k)} - \eta \nabla f(\boldsymbol{x}^{(k)}))$$

3. 线搜索步长梯度下降： 示例



$$\eta = 0.01, f_{\min} = 4.69$$

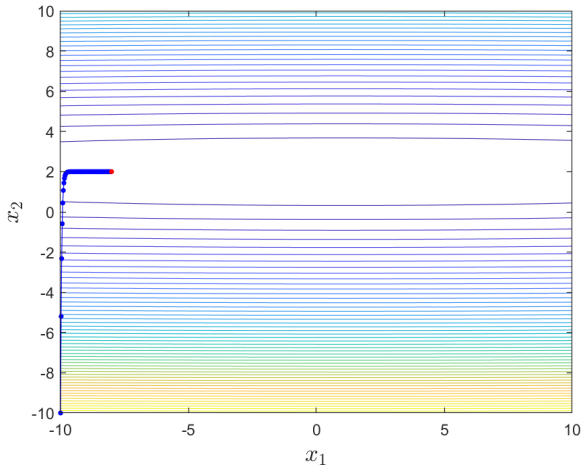
梯度下降



$$\eta = \eta^*, f_{\min} = 2.92$$

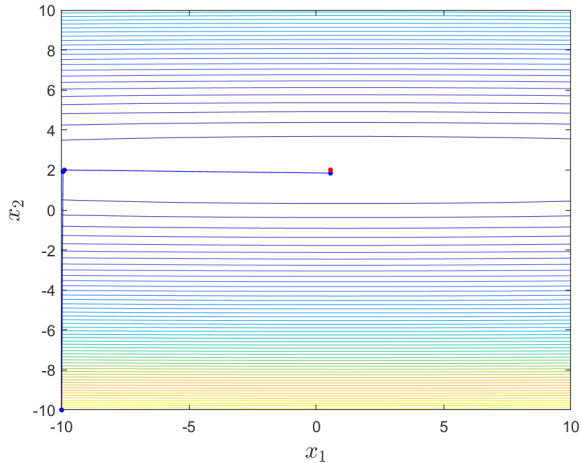
线搜索梯度下降

3. 线搜索步长梯度下降： 示例



$$\eta = 0.1, f_{\min} = 0.81$$

梯度下降



$$\eta = \eta^*, f_{\min} = 0.00$$

线搜索梯度下降